

Archana Avancha Data Analyst

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Profile

Results-oriented Data Analyst with 2+ years of experience in statistical analysis and data modeling. Proficient in SQL and Python for data manipulation and analytics, including predictive modeling and machine learning. Skilled in using Tableau and Power BI for creating dynamic dashboards and reports. Experienced in agile methodologies like Scrum and Kanban, and uniting with cross-functional teams to drive data-driven decisions and improve operational efficiency.

Skills

Programming Languages: Python, SQL, NoSQL, HTML, CSS

Frameworks & Tools: NumPy, Pandas, Matplotlib, GitHub, Docker, CI/CD, Statistics, Agile, Hadoop, Hive, PySpark, Snowflake, Seaborn, Sci-kit Learn, Jira, KPI Analysis, Automation Power Query, TensorFlow, PyTorch, OpenCV, Multi-threading, Advanced Analytics, Apache

Databases: MySQL, MS-SQL, Redshift, Big Query, PostgreSQL, MongoDB, SparkSQL, DynamoDB, RDBMS

Data Analysis & ETL Tools: Power BI, Tableau, Data Bricks, MS Excel, Google Analytics, Data (Modeling, Mapping, Mining, Extraction, Transformation), Salesforce, Zendesk, Confluence, Postman, SAP ERP, ETL, Data Processing, Data Wrangling, Data Pipelines

Professional Experience

Data Analyst, Hitachi

09/2023 – Present | Remote, USA

- Moulded a real-time supply chain analytics dashboard using SQL to query data from multiple sources and Python to process and analyze, creating a scalable solution to monitor key metrics like delivery times and inventory levels.
- Leveraged Pandas to handle over 1 million rows of data, NumPy for performing numerical calculations, and Matplotlib to create over 20 types of charts, this tools supported detailed analysis and visualization of supply chain metrics.
- Controlled data with SQL by creating 50+ tables, optimizing performance with 20+ indexes, and combining data with 10+ joins, applying normalization to reduce data redundancy and improve query efficiency.
- Gathered data using SQL and integrated it into Azure Synapse Analytics, processing up to 5 TB of data.
- Used Azure Synapse's cloud-based architecture for storage and efficient ETL processes, ensuring timely data availability.
- Operated Azure Functions for serverless data processing, handling up to 200,000 requests per month
- Monitored performance with Azure Monitor, configuring 50+ alerts for real-time accuracy and system reliability.
- Prepared interactive dashboards and reports with Power BI, integrating data from 10+ sources, and creating 15+ dynamic visualizations with filters and slicers to provide actionable insights for supply chain optimization.

Data Analyst, AgFirst Farm Credit Bank

05/2023 – 08/2023 | Columbia, USA

- Directed a critical migration from a legacy Mainframe to a new DNA data server, driving the transition of customer, ACA, and loan records worth over \$750 million for accurate and seamless data migration throughout the project.
- Utilized Python libraries like Pandas and NumPy for refined data manipulation and analysis, automating data extraction and processing tasks to streamline workflows and complex analytics during the migration process.
- Executed over 500 SQL queries for data extraction and validation, using statistical methods for accuracy and gaining a 95% validation rate on the first pass, to reduce the project timeline by 20% through optimized methodologies.
- Applied SAS software for in-depth data analysis, including statistical checks and data quality checks, uncovering valuable insights for high data integrity to helped effective decision-making.
- Designed interactive dashboards using PowerBI to monitor migration progress and data integrity, creating visualizations that enabled stakeholders to evaluate impact over 10,000 ACA records post-migration

Data Analyst, Cognizant Technologies

02/2022 – 08/2022 | Hyderabad, India

- Performed comprehensive data analysis on aviation industry metrics such as flight schedules, passenger traffic, aircraft maintenance records, fuel consumption, and safety incidents using Python.
- Integrated weather information, airspace restrictions, and airport congestion data with internal aviation data using Apache Spark and Apache Kafka to enhance situational awareness .
- Leveraged AWS Lambda serverless computing platform to build event-driven microservices for handling real-time data processing, integrating third-party APIs within the aviation sector.
- Applied machine learning algorithms, specifically Random Forest Regression and Gradient Boosting models, to historical aviation data to forecast demand for flights and optimize pricing structures.
- Designed and maintained MySQL databases to support the storage, organization, and retrieval of crucial aviation data, such as flight schedules, passenger information, and aircraft details.

Education

Master of Science, University of North Carolina
Computer Science

08/2022 – 12/2023 | Charlotte, USA

Bachelor of Technology, Neil Gogte Institute of Technology
Computer Science

08/2018 – 05/2022 | Hyderabad, India